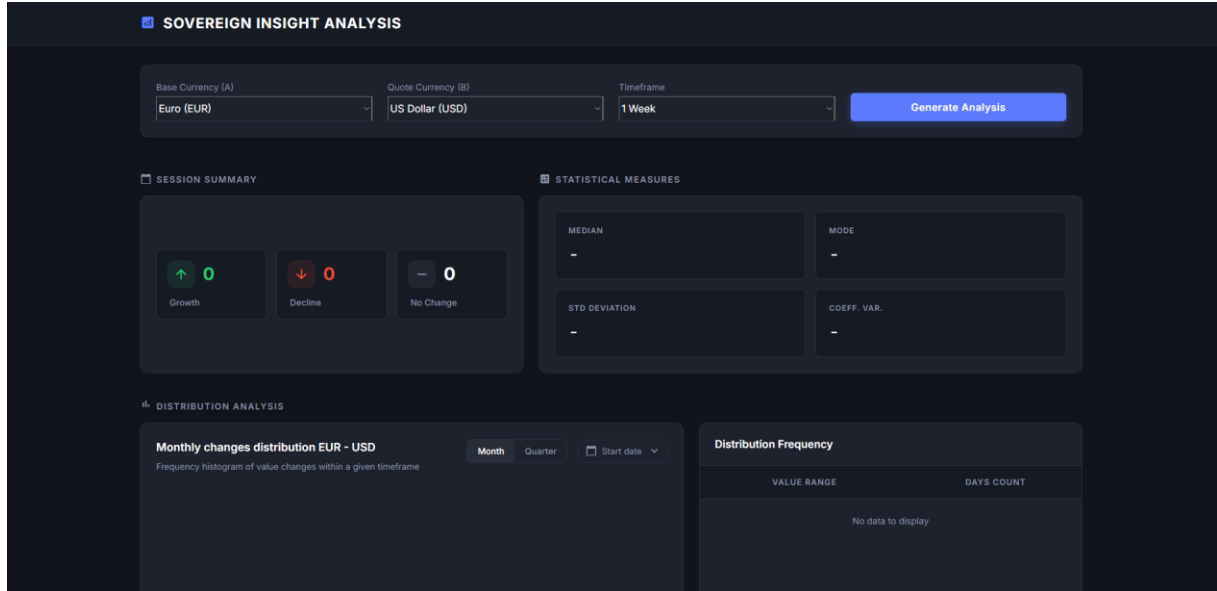
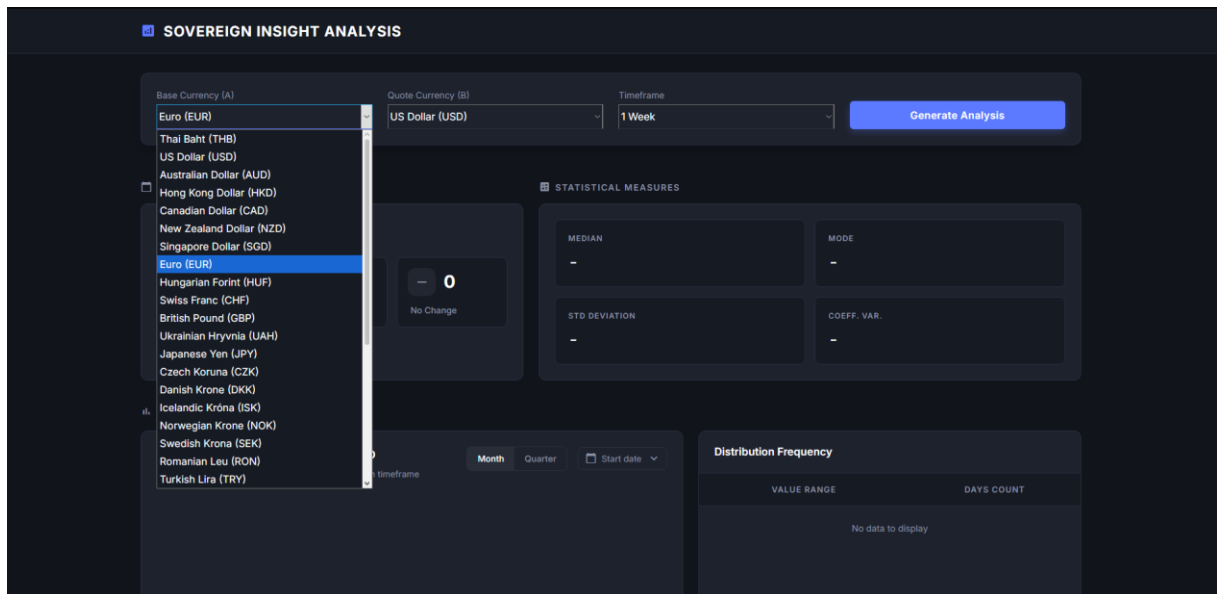


Step 1: Configure and Generate Currency Analysis

Access the Application: Open the web application in your browser, ensuring you have an active internet connection to communicate directly with the external API.

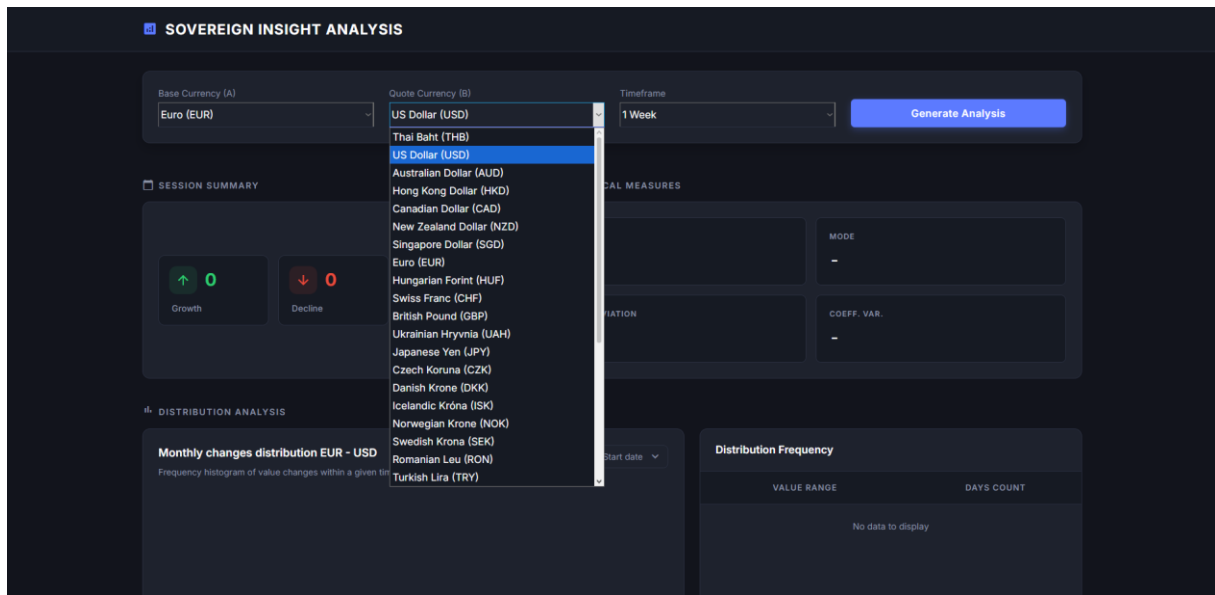


Select Base Currency: Navigate to the top navigation bar (Header & Global Configuration) and use the first dropdown menu to select your desired Base Currency. The system supports all major currencies available in the NBP average exchange rate table, such as USD, EUR, GBP, and CHF.

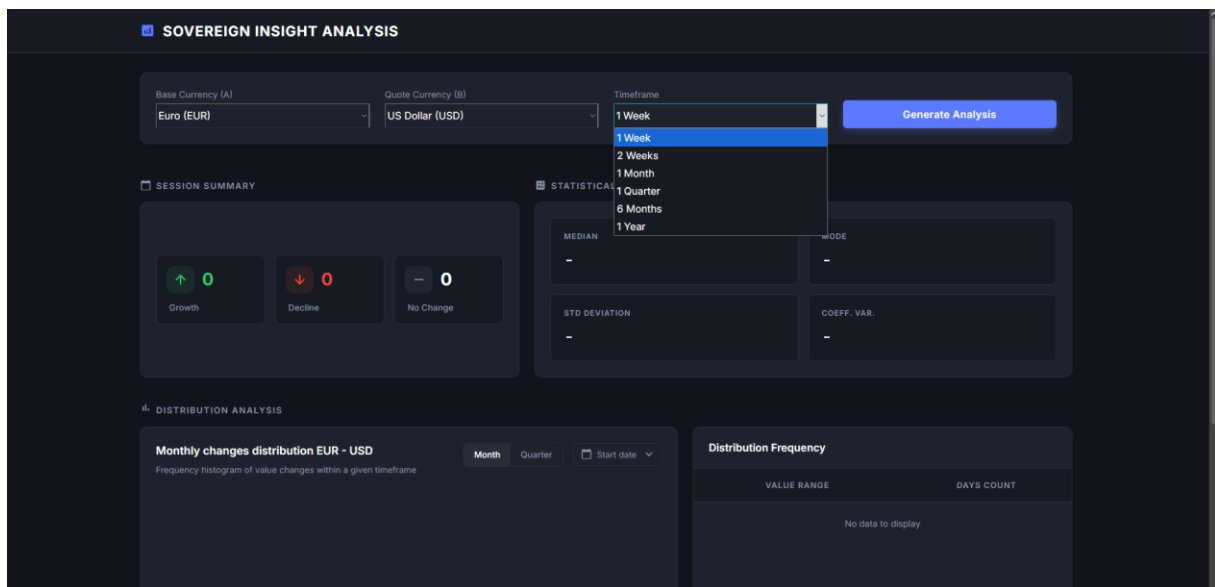


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Select Quote Currency: Use the second dropdown menu to select a Quote Currency if you wish to compare a currency pair or evaluate a synthetic cross-rate.



Choose Timeframe: Use the global Timeframe Control selector to select a predefined historical analysis period. You can choose between **1 Week**, **2 Weeks**, **1 Month**, **1 Quarter**, **6 Months**, or **1 Year**.



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Run the Analysis: Click the **"Generate Analysis"** button to validate your inputs and initiate the data retrieval.

The screenshot displays the 'SOVEREIGN INSIGHT ANALYSIS' application interface. At the top, there are three input fields: 'Base Currency (A)' set to 'Euro (EUR)', 'Quote Currency (B)' set to 'US Dollar (USD)', and 'Timeframe' set to '1 Week'. A blue button labeled 'Generate Analysis' is positioned to the right of these fields, with a blue arrow pointing to it. Below the input fields, the interface is divided into three main sections: 'SESSION SUMMARY', 'STATISTICAL MEASURES', and 'DISTRIBUTION ANALYSIS'. The 'SESSION SUMMARY' section shows three indicators: 'Growth' with a green up arrow and '0', 'Decline' with a red down arrow and '0', and 'No Change' with a grey dash and '0'. The 'STATISTICAL MEASURES' section displays four metrics: 'MEDIAN' with a dash, 'MODE' with a dash, 'STD DEVIATION' with a dash, and 'COEFF. VAR.' with a dash. The 'DISTRIBUTION ANALYSIS' section is further divided into 'Monthly changes distribution EUR - USD' and 'Distribution Frequency'. The 'Monthly changes distribution EUR - USD' section has a 'Month' tab selected and a 'Start date' dropdown. The 'Distribution Frequency' section shows a table with columns 'VALUE RANGE' and 'DAYS COUNT', but it contains the text 'No data to display'.

- **Automated Background Processing:** The application will automatically fetch data from the official National Bank of Poland (NBP) Web API. If your selected timeframe exceeds the NBP API limit of 93 days (e.g., 6 months or 1 year), the client-side system will automatically split the request into multiple smaller chunks, execute parallel asynchronous requests, and merge them into a single chronological dataset.

- **Data Sanitization:** The system automatically filters out non-trading days (weekends and national holidays), processing only valid trading data points. Visual feedback like skeleton loaders or pulse animations will display during loading times.

Handle Potential Errors:

- If the external NBP API is down, the system will display a user-friendly error message advising you to try again later.

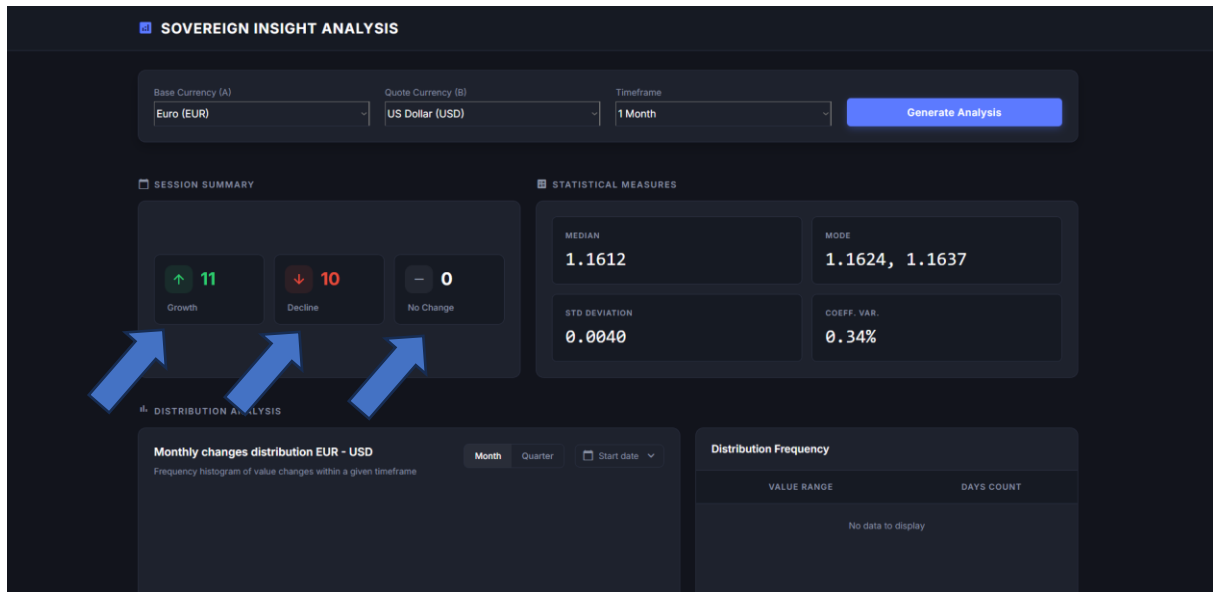
- If your selected timeframe consists entirely of non-trading days (e.g., a single weekend), the system will alert you that no data is available for that period.

Step 2: Evaluate Statistical Measures & Session Trends

Once the application state updates and data is successfully loaded, review the synchronized dashboard metrics:

Analyze Session Trends: Look at the **Session Summary Cards** panel to view the market's directional momentum. Read the precise calculated counts of:

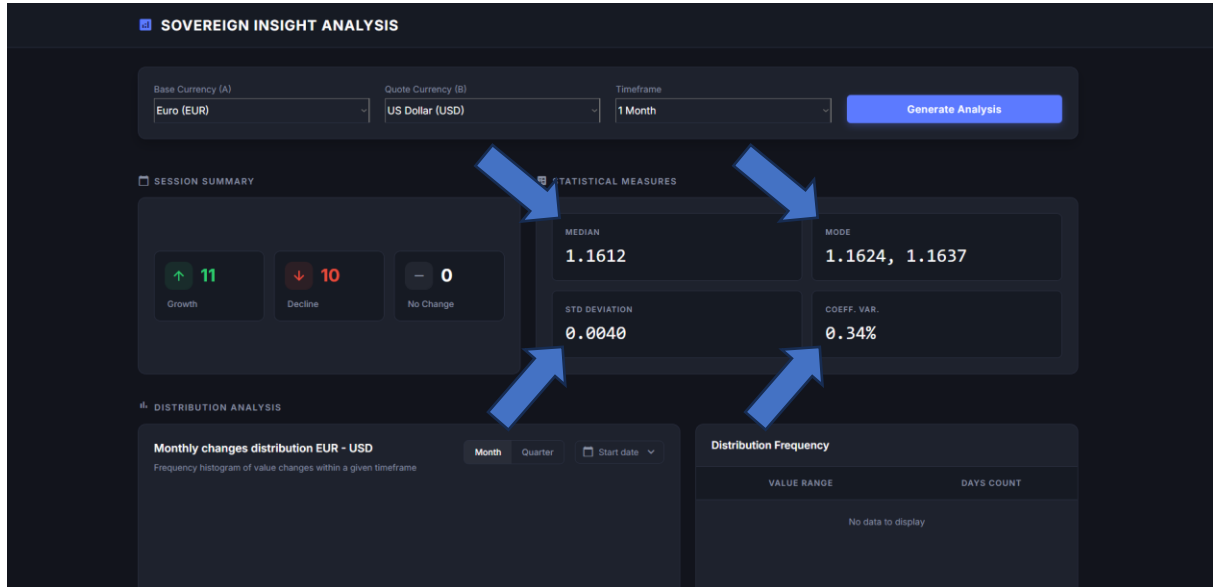
- **Upward** (growth) sessions.
- **Downward** (decline) sessions.
- **No-change** (stable/unchanged) sessions.



SOVEREIGN INSIGHT ANALYSIS – USER MANUAL

Review Advanced Statistics: Examine the **Statistical Measures Grid**, a structured 2x2 layout displaying real-time calculations from the client-side mathematical engine:

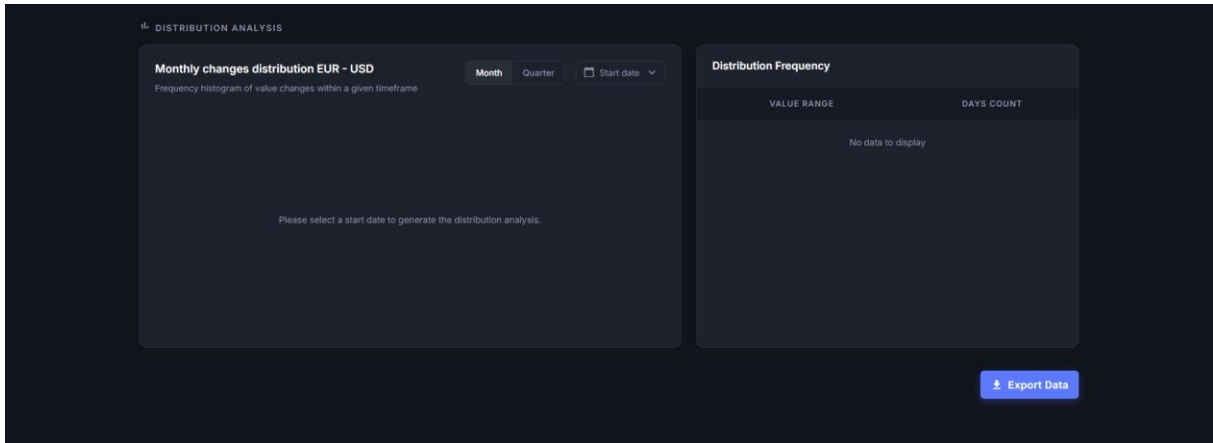
- **Median:** Identifies the central tendency of the exchange rate over the selected timeframe.
- **Mode:** Displays the most frequent value within the exchange rate dataset.
- **Standard Deviation:** Quantifies absolute market volatility and data dispersion.
- **Coefficient of Variation (Coeff. Var):** Measures relative market risk, expressed explicitly as a percentage.



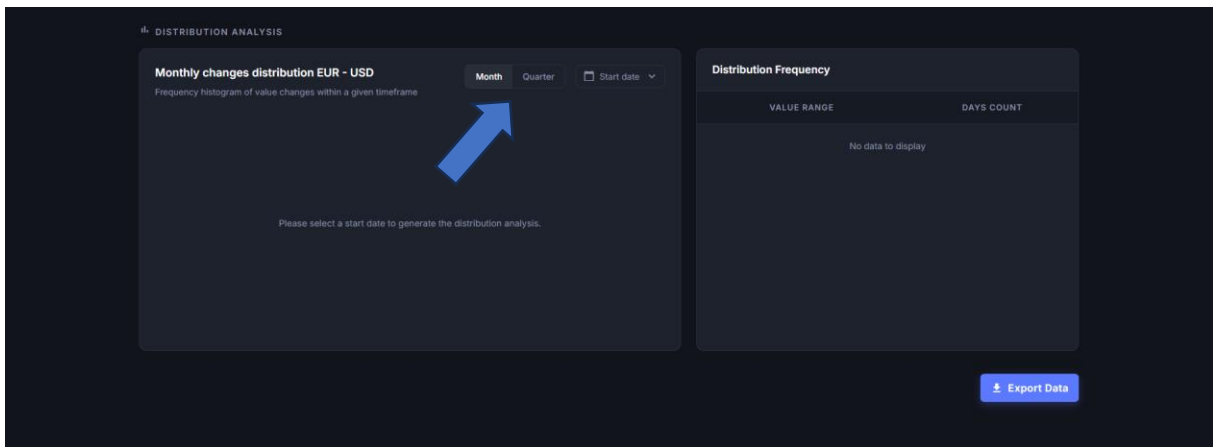
Step 3: Analyze Change Distribution

To investigate how exchange rate fluctuations are distributed over time between consecutive trading sessions, proceed to the bottom dashboard module:

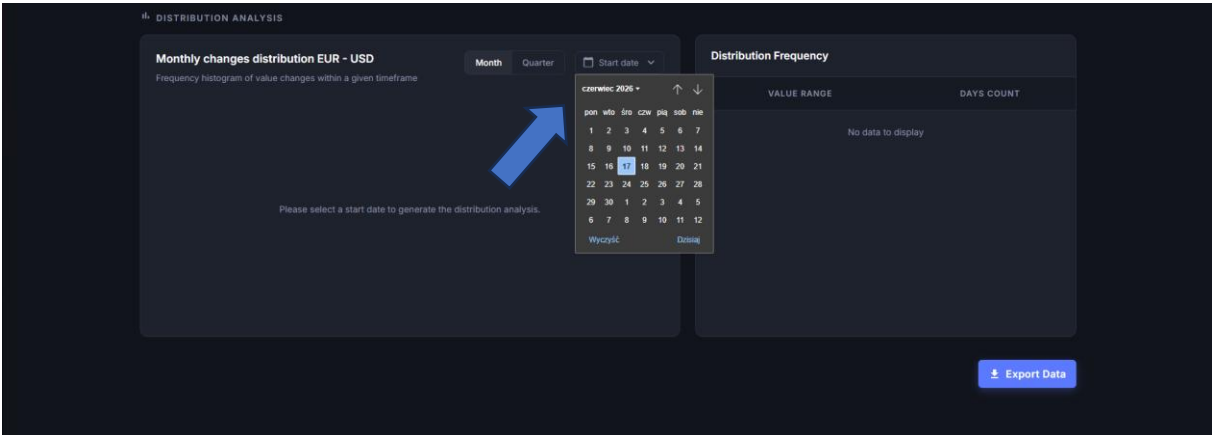
Navigate: Scroll down to the **"Change Distribution Analysis"** section.



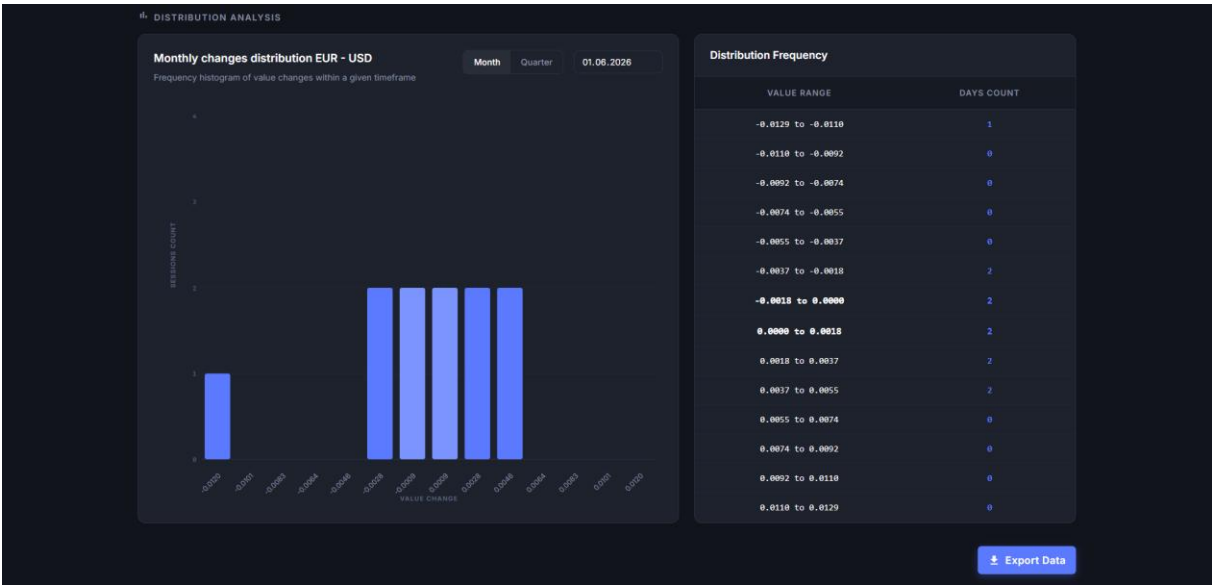
Set Aggregation Mode: Click the **Aggregation Toggle** buttons to choose how you want to group the historical fluctuations. You can select either **"Month"** or **"Quarter"** mode.



Select Start Date: Click the **Start Date Picker** (calendar interface) to explicitly define the exact starting point in history for this distribution analysis.



Interpret the Dynamic Histogram: The system calculates the percentage differences between consecutive sessions, groups them into frequency bins, and renders a visual histogram using the Recharts library, showcasing a pyramid-like structure of market volatility. **Read Raw Data Tables:** Look at the adjacent **Frequency Results Table** to view the structured text and precise numerical session counts mapped directly to each specific value range.

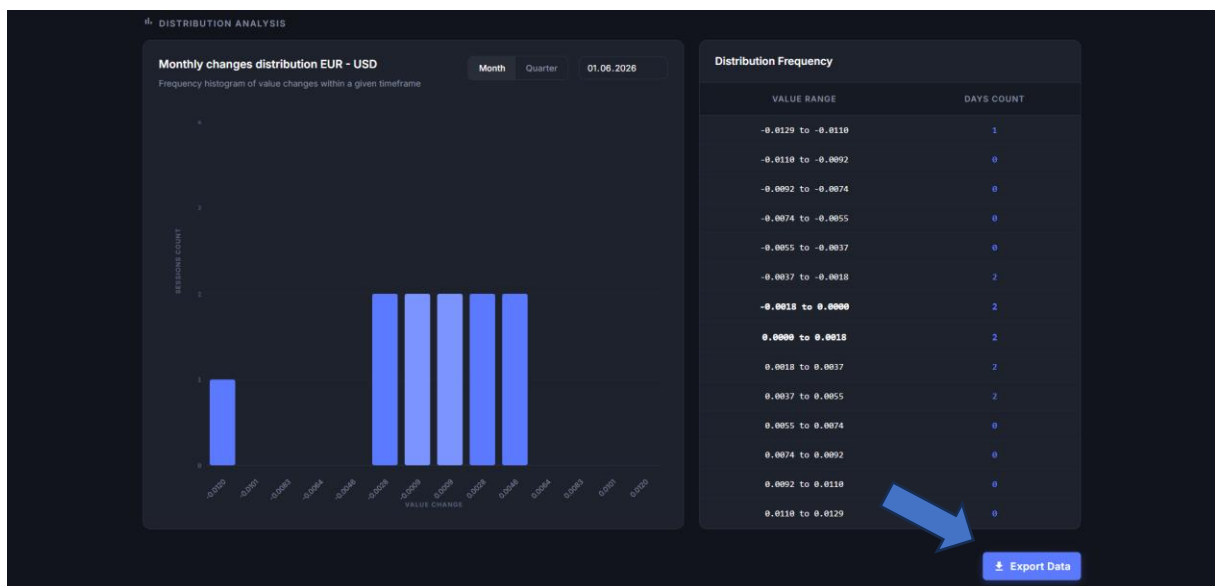


Step 4: Export Analysis Results and Visualizations

To save your data or charts for external reporting, use the export utility built entirely into the client side:

Locate the Trigger: Find the **"Export Data"** button located in the bottom-right corner of the interface.

- *Note:* If the data is still loading or has not been generated yet, this button will remain disabled to prevent downloading corrupted or blank files.



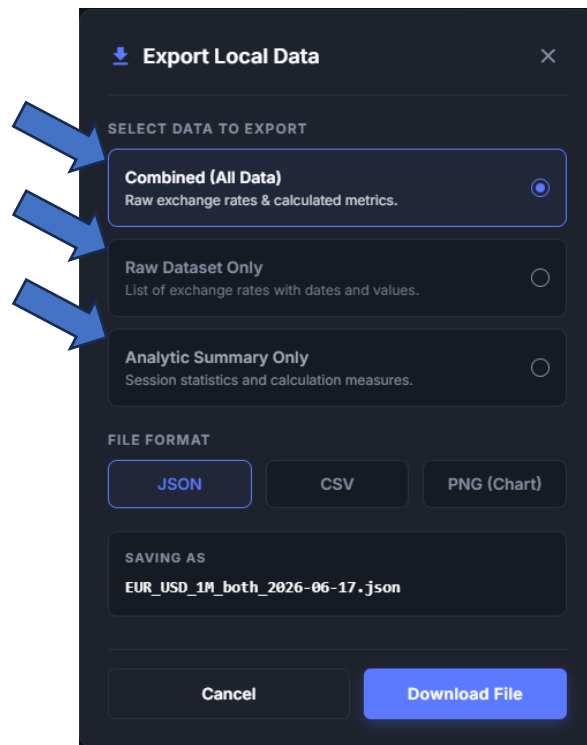
Open the Export Menu: Click the active **"Export Data"** button to open the **Export Modal** overlay.

The 'Export Local Data' modal is shown with the following options:

- SELECT DATA TO EXPORT:**
 - Combined (All Data)** (Selected): Raw exchange rates & calculated metrics.
 - Raw Dataset Only: List of exchange rates with dates and values.
 - Analytic Summary Only: Session statistics and calculation measures.
- FILE FORMAT:**
 - JSON** (Selected)
 - CSV
 - PNG (Chart)
- SAVING AS:** EUR_USD_1M_both_2026-06-17.json
- Buttons:** Cancel, Download File

Select Data Scope: Choose the depth of your export file using the radio buttons:

- **Combined (All Data):** Packages the complete raw historical dataset alongside all computed statistical metrics.
- **Raw Dataset Only:** Isolates and exports only chronological exchange rates and calendar dates, omitting statistics.
- **Analytic Summary Only:** Extracts only calculated session trends and advanced statistical measures, omitting daily logs.



The screenshot shows a dark-themed dialog box titled "Export Local Data" with a close button (X) in the top right corner. Below the title bar, there is a section labeled "SELECT DATA TO EXPORT" containing three radio button options:

- Combined (All Data)**: Raw exchange rates & calculated metrics. (Selected, indicated by a filled blue circle)
- Raw Dataset Only**: List of exchange rates with dates and values. (Unselected, indicated by an empty circle)
- Analytic Summary Only**: Session statistics and calculation measures. (Unselected, indicated by an empty circle)

Below these options is a section labeled "FILE FORMAT" with three buttons: "JSON" (highlighted with a blue border), "CSV", and "PNG (Chart)".

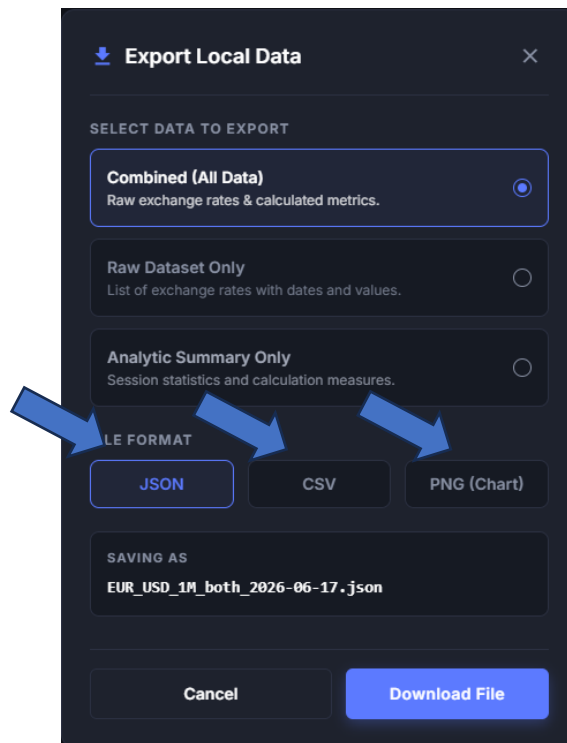
At the bottom of the dialog, there is a "SAVING AS" section with a text input field containing the filename "EUR_USD_1M_both_2026-06-17.json".

At the very bottom, there are two buttons: "Cancel" and "Download File" (highlighted in blue).

Three blue arrows are overlaid on the left side of the dialog, pointing to the "SELECT DATA TO EXPORT" section, specifically towards the "Combined (All Data)" option.

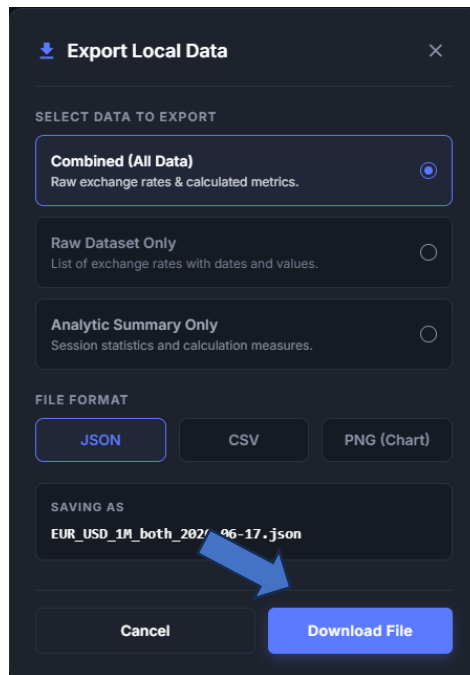
Select Your Preferred Format: Click on one of the available download options depending on your analytical needs:

- **CSV:** Parses and downloads the raw exchange rates and session frequency tables into an editable spreadsheet file.
- **JSON:** Generates a structured data file containing the complete raw session state and computed metrics.
- **PNG / JPEG:** Captures the rendered Recharts distribution histogram, converting the visual SVG components into a high-resolution standalone image file.



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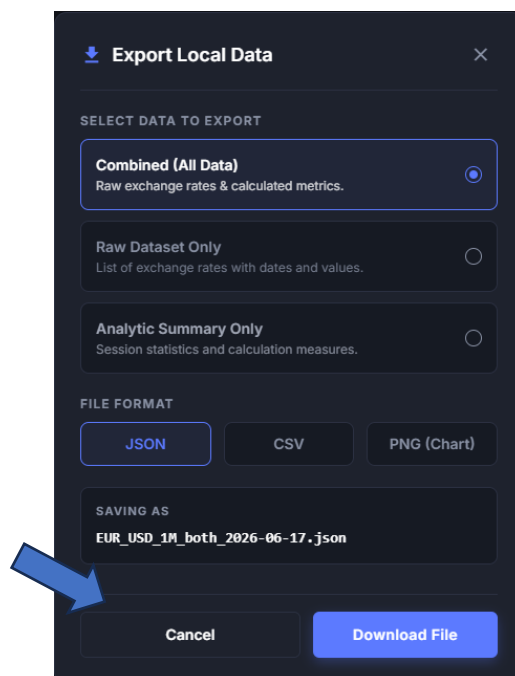
Download Completion: The system automatically compiles the asset, creates a temporary download link, and prompts your web browser to save the file (e.g., analysis_EUR_USD.csv or histogram_report.png) onto your local device.



The screenshot shows the 'Export Local Data' modal with the following elements:

- SELECT DATA TO EXPORT:** Three radio button options: 'Combined (All Data)' (selected), 'Raw Dataset Only', and 'Analytic Summary Only'.
- FILE FORMAT:** Three buttons: 'JSON' (selected), 'CSV', and 'PNG (Chart)'.
- SAVING AS:** A text field displaying 'EUR_USD_1M_both_2024-06-17.json'.
- Buttons:** 'Cancel' and 'Download File' (highlighted with a blue arrow).

Cancel Export: If you change your mind, you can close the modal menu without selecting a format, and the system will safely return you to the dashboard view with no further actions taken.



The screenshot shows the 'Export Local Data' modal with the following elements:

- SELECT DATA TO EXPORT:** Three radio button options: 'Combined (All Data)' (selected), 'Raw Dataset Only', and 'Analytic Summary Only'.
- FILE FORMAT:** Three buttons: 'JSON' (selected), 'CSV', and 'PNG (Chart)'.
- SAVING AS:** A text field displaying 'EUR_USD_1M_both_2026-06-17.json'.
- Buttons:** 'Cancel' (highlighted with a blue arrow) and 'Download File'.